

- 3 (a) (i) The kinetic theory of gases leads to the equation

$$\frac{1}{2} m \langle c^2 \rangle = \frac{3}{2} kT.$$

Explain the significance of the quantity $\frac{1}{2} m \langle c^2 \rangle$.

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- (ii) Use the equation to suggest what is meant by the absolute zero of temperature.

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[3]

- (b) Two insulated gas cylinders **A** and **B** are connected by a tube of negligible volume, as shown in Fig. 3.1.

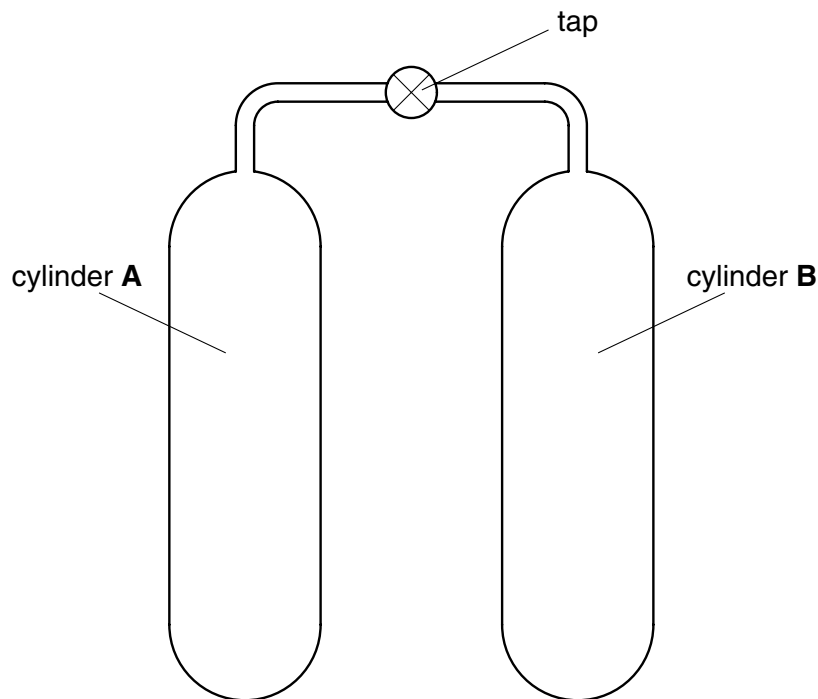


Fig. 3.1

Each cylinder has an internal volume of $2.0 \times 10^{-2} \text{ m}^3$. Initially, the tap is closed and cylinder **A** contains 1.2 mol of an ideal gas at a temperature of 37°C . Cylinder **B** contains the same ideal gas at pressure $1.2 \times 10^5 \text{ Pa}$ and temperature 37°C .

- (i) Calculate the amount, in mol, of the gas in cylinder **B**.

amount = mol

- (ii) The tap is opened and some gas flows from cylinder **A** to cylinder **B**. Using the fact that the total amount of gas is constant, determine the final pressure of the gas in the cylinders.

pressure = Pa
[6]